

T-Sheehan

# Zero Sector

Game Design Document

Thomas Sheehan  
3-18-2026

# Zero Sector

## Game Overview

**Title:** Zero Sector

**Genre:** 2D Top-Down Shooter

**Platform:** PC

**Engine:** Unity

**Target Audience:** Fans of fast-paced arcade shooters and modern roguelikes (e.g., *Bosconian*, *Geometry Wars*, *Hades*)

**Game Summary:** *Zero Sector* is a fast-paced top-down space shooter where players navigate confined sectors, destroy enemy forces, and survive escalating threats using skill-based movement, combat mechanics, and strategic upgrades.

The game emphasizes:

- Precision movement and dodging
- Reactive combat (perfect dodge system)
- Build customization through upgrades
- Replayability via procedural sector progression

## Core Gameplay

1. Enter a **procedurally generated sector**
2. Engage enemies and avoid incoming fire
3. Build combo through successful hits
4. Earn **run currency**
5. Clear sector objectives (eliminate enemies/bases)
6. Transition to next sector
7. Repeat until:
  - Player death (permadeath)
  - Boss milestones reached

# Core Mechanics

## Movement & Combat

- 360° movement
- Directional shooting (mouse or twin-stick)
- Dodge mechanic with invulnerability window

## Reactive Combat (Perfect Dodge System)

### Description:

A skill-based mechanic rewarding precise timing.

### Mechanics:

- Dodge within a short window before impact
- Triggers:
  - Brief slow-motion effect
  - Special counterattack
  - Temporary invulnerability

### Example Effects (upgrade-dependent):

- Shockwave blast (AoE)
- Chain lightning
- EMP pulse
- Missile barrage

## Combat Systems

### Primary Weapon

- Continuous fire with cooldown
- Builds combo meter

### Combo System

- Builds with consecutive hits
- Resets on:
  - Taking damage

- Missing shots (tunable)

**Effects:**

- Increased fire rate
- Increased damage (optional scaling)
- Visual/audio feedback

**Secondary Weapon (Bomb)**

- Clears projectiles + enemies in radius
- Recharge via pickups or rewards

**Special Ability System**

- Cooldown-based abilities
- High-impact effects

**Examples:**

- EMP blast
- Black hole pull
- Overdrive mode

**Decoy System**

- Deploys temporary AI allies

**Behavior:**

- Attract enemy fire
- Deal damage (lower than player)
- Block projectiles

**Variants:**

- Attack decoy
- Shield decoy
- Explosive decoy

# Game Systems

## Game States

- Main Menu
- Playing
- Paused (optional)
- Game Over

## Game Loop

- Input → Movement → Shooting → Collision → State Check

# Player Design

## Abilities:

- Smooth directional movement
- Instant shooting (with cooldown)
- Upgradeable Ship/Weapons
  - Fire patterns (machine gun, sniper, homing, burst, pistol, ricochet, Charge shot)
  - Elemental Effects (plasma burn, EMP, Explosive)
  - Defensible upgrades (energy shield, emergency barrier, phase shift, armour)
  - Mobility upgrades (Afterburner, dash, drift control, gravity sling, blink)
  - Drones (Decoys, temporary, attack, defend, repair)
- Special abilities (Time slow, overdrive, combo meter, emp blast)
- Synergy Upgrades (vampiric shots, chain lightning, last stand)

## Constraints:

- Limited health
- No infinite firing

# Enemy Design

| Enemy Type     | Behavior                                                                       | Role in Gameplay                                                  | Threat Level |
|----------------|--------------------------------------------------------------------------------|-------------------------------------------------------------------|--------------|
| Interceptor    | Aggressively chases the player in real-time, often in small groups             | Core pressure unit; forces constant movement and dodging          | Medium       |
| Swarm Units    | Spawn in bursts and rush the player quickly with low health                    | Overwhelm player; punishes slow reaction and poor positioning     | Medium       |
| Base Turret    | Stationary, attached to enemy bases, fires at player within range              | Area denial; protects objectives and limits safe zones            | Low–Medium   |
| Commander Unit | Stays near bases and enhances nearby enemies (speed, aggression, or fire rate) | Priority target: killing it weakens the entire sector             | High         |
| Defense Drone  | Orbits around bases or commanders, blocking shots or intercepting player       | Defensive layer; forces repositioning and precision aiming        | Medium       |
| Mine Layer     | Moves slowly and drops explosive mines behind or around the player             | Area control: restricts movement and punishes careless navigation | Medium       |
| Sniper Unit    | Keeps distance and fires slow, high-damage precision shots                     | Long-range threat; forces awareness and prioritization            | High         |
| Boss Unit      | Large enemy with unique attack patterns (multi-phase optional)                 | Sector milestone; tests player mastery and build                  | Very High    |

# Boss Design

Bosses appear at milestones (Ex, every 10 waves)

## Phases:

- Phase 1: Basic patterns
- Phase 2: Increased aggression (new attacks, faster)
- Phase 3: Desperation Phase (1 big attack)

## Design Purpose:

- Break pacing
- Player skill check
- Add progression milestones

Which boss player faces can change depending on certain in game stats, and where the player is in progression (i.e. player has lower health, boss might be less bulky or fast. The further in a run the player is, the harder the boss will be, etc.)

# Level Design

## Structure:

- Game divided into procedural sectors (arena-style)
- Each sector includes:
  - Enemy spawns
  - Obstacles/hazards
  - Bases or objectives

## Flow:

- Enter sector
- Assess threats
- Destroy targets
- Survive waves
- Warp to next sector

# Progression Systems

## Dual Currency:

- Run Currency (Earned during run, used in safe zones, Lost on death)
- Meta Currency (Earned based on performance, Used for permanent upgrades)

## Upgrade Categories:

- Weapons (Spread, piercing, explosive, homing, charge)
- Defense (Shields, armour, mobility, abilities, synergies)

# Death & Replayability

Permadeath system encourages repeated runs and build experimentation

# Game States

- Main Menu
- Playing
- Paused
- Game Over

# Visual Design

- Top-down perspective
- Clean arcade-style visuals
- Bright projectiles and effects
- High Readability



# Audio Design

## SFX:

- Shooting
- Explosion
- Hit Feedback
- Combo Feedback

## Music:

- Fast paced electronic / arcade style
- Intensity scales with gameplay

# Technical Design

## Core Systems:

- PlayerController
- EnemyAI
- Projectile system
- GameManager
- Spawner system

## Tech Features:

- Unity 2D physics
- Collision detection
- State management
- Object pooling (future optimization)

# Procedural Generation Design

## Overview:

*Zero Sector* uses procedural generation to create replayable combat scenarios through dynamically constructed sectors.

Rather than generating infinite open space, the game generates contained combat areas ("Sectors") with controlled enemy compositions and objectives, and semi-randomized layouts.

## Sector Generation Approach:

Each sector is generated using a modular system, combining:

- Predefined enemy spawn configurations
- Hazard placement (asteroids, planets, environmental obstacles)
- Objective types

## Generation Elements:

- Area Layout
  - Selected from a pool of predesigned templates
  - Ensures gameplay clarity and balance
- Enemy Compositions
  - Enemy pools based on sector objectives or theme (*Future expansion*)
  - Mix of enemy roles (pressure, defense, ranged, support)
- Spawn Patterns
  - Timed waves or continuous spawning
  - Scales with progression
- Difficulty Scaling
  - Increased enemy count
  - Faster behaviours
  - Introduction of Elite variants (enhanced stats and new attack patterns)

## Scope Considerations:

For initial implementation, procedural generation will be limited to a small pool of sector designs and basic enemy randomization to ensure feasibility within project constraints. (Subject to refinement or reconsideration based on development progress)

## Design Intent:

The system prioritizes:

- Replayability through variation
- Consistency through controlled templates
- Performance through simple generation logic

## Testing Plan

- Test movement responsiveness
- Test shooting accuracy
- Test collision reliability
- Test Difficulty scaling

## Future Features

- Power Ups
- Boss Enemies
- Score System + Leaderboard
- Soundtrack Expansion
- Visual Polish (particles, screen shake, etc.)

# Design Justification

## Core Design Philosophy:

Zero Sector is designed to merge the immediacy of classic arcade shooters with the long-term engagement of modern roguelikes. The goal is to create a game that is easy to pick up, hard to master, and highly replayable.

## Skill Based Gameplay Focus:

Unlike many roguelikes that rely heavily on stat progression, Zero Sector emphasizes the player skill as the primary driver of success. This is achieved through precision movement mechanics, enemy pressure, and the perfect dodge mechanic which rewards awareness and timing over passive upgrades.

## Replayability Through Systems, Not Content:

Instead of relying on handcrafted levels, Replayability is driven by procedurally generated sectors, upgrade combinations, and dynamic enemy compositions, ensuring each run feels different without requiring massive asset production.

## Build Diversity:

Players can customize their playstyles through weapon modifications, ability choices, and synergy systems. Like high-speed glass canon builds, defensive/survivability builds, or ability focused builds. This lets the player experiment with multiple playstyles on every new run.

## Combat Readability and Clarity:

Given the fast-paced nature of the gameplay, visual clarity must be prioritized. Clear enemy silhouettes, distinct projectile colours, and minimal visual clutter to ensure players can make fast, informed decisions during the chaotic combat.

## Risk/Reward Systems:

Mechanics like the combo meter and perfect dodge encourage the player to take risks for big rewards. This is to create tension and engagement in each run during every encounter, while preventing a passive "AFK" approach.

## Modernizing a Classic:

Inspired originally by Bosconian, the game evolves the formula by introducing roguelike mechanics, ability systems, Dynamic boss encounters, and meta progression. The intent is to respect the original design while making it relevant to modern players.